

2(a)	<i>two from:</i> 1 the number of times traps used ; 2 the type of trap used ; 3 time (of day) moths were trapped ; 4 time of year moths were trapped ; 5 ref. to positioning / spacing of traps ; 6 number of traps used ; 7 size of area from which samples taken ; 8 method of counting ;	2
2(b)	<i>three from:</i> <i>description:</i> 1 (melanic moths) increase in frequency more in area X than in area Y ; 2 melanic moths in area X increase, most rapidly/linearly, and then starts to slow and Y increases more slowly at first and then increases more rapidly ; <i>explanation:</i> 3 area X more polluted than Y so selection acts more strongly AW / ORA ; 4 some non-melanics remain in population X because of breeding between heterozygotes ;	3

Question	Answer	Marks
2(c)	line starting from generation 10 and below X and above Y and to 24 generations ;	1
2(d)	<i>two from:</i> <i>melanic forms/they:</i> 1 less predated by species other than birds/named likely predator ; 2 less susceptible / (more) resistant, to poisoning by toxins / pollutants ; 3 higher fitness / produce more offspring ; 4 more resistant to disease ; 5 better at competing with, new / alien / introduced species ; 6 reference to climate change ;	2

Question	Answer	Marks																
2(e)(i)	<table><tr><th>category</th><th>O</th><th>E</th><th>$\frac{(O - E)^2}{E}$</th></tr><tr><td>melanic</td><td>56</td><td>52</td><td>0.31</td></tr><tr><td>non-melanic</td><td>48</td><td>52</td><td>0.31</td></tr><tr><td colspan="3">$\chi^2 =$</td><td>0.62</td></tr></table> correct expected numbers, <u>52 and 52</u> ; correct values for $(O - E)^2 / E$; correct values for χ^2 ;	category	O	E	$\frac{(O - E)^2}{E}$	melanic	56	52	0.31	non-melanic	48	52	0.31	$\chi^2 =$			0.62	3
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2(e)(ii)	difference between expected and observed is not significant because the value for chi-squared is less than the critical value at p = 0.05 / 5% or p = 0.10 / 10% ;	1																